

PL5 — Broken-Wing Put Butterfly (SPX)

Presentation report · Prop Desk Quant · 2026-06

QuantConnect backtest · Jun 2021 - Jun 2026 (5 years) · enters every Friday 10:00 ET

WHAT IT IS — a defined-risk put structure with a crash-convex tail.

Per unit: +1 put @ -30 delta / -2 puts @ -18 delta / +2 puts @ -3 delta.

Small net debit. Profit 'tent' if the market drifts down moderately; defined max loss in a valley near the lower strike; the position turns convex again far below (the crash tail).

WHAT WE TESTED — 4 holding horizons: 21 / 28 / 45 / 60 DTE, each over the full 5 years.

KEY TAKEAWAYS

1. The result is DECIDED BY EXECUTION, not by the structure being right or wrong.

Same backtest, 28 DTE: \$+131,102 at mid -> \$+4,087 if every leg is crossed at full bid/ask. The 5-leg entry spread is what moves the number.

2. Traded properly (as ONE combo at a limit price), realistic fill is close to mid:

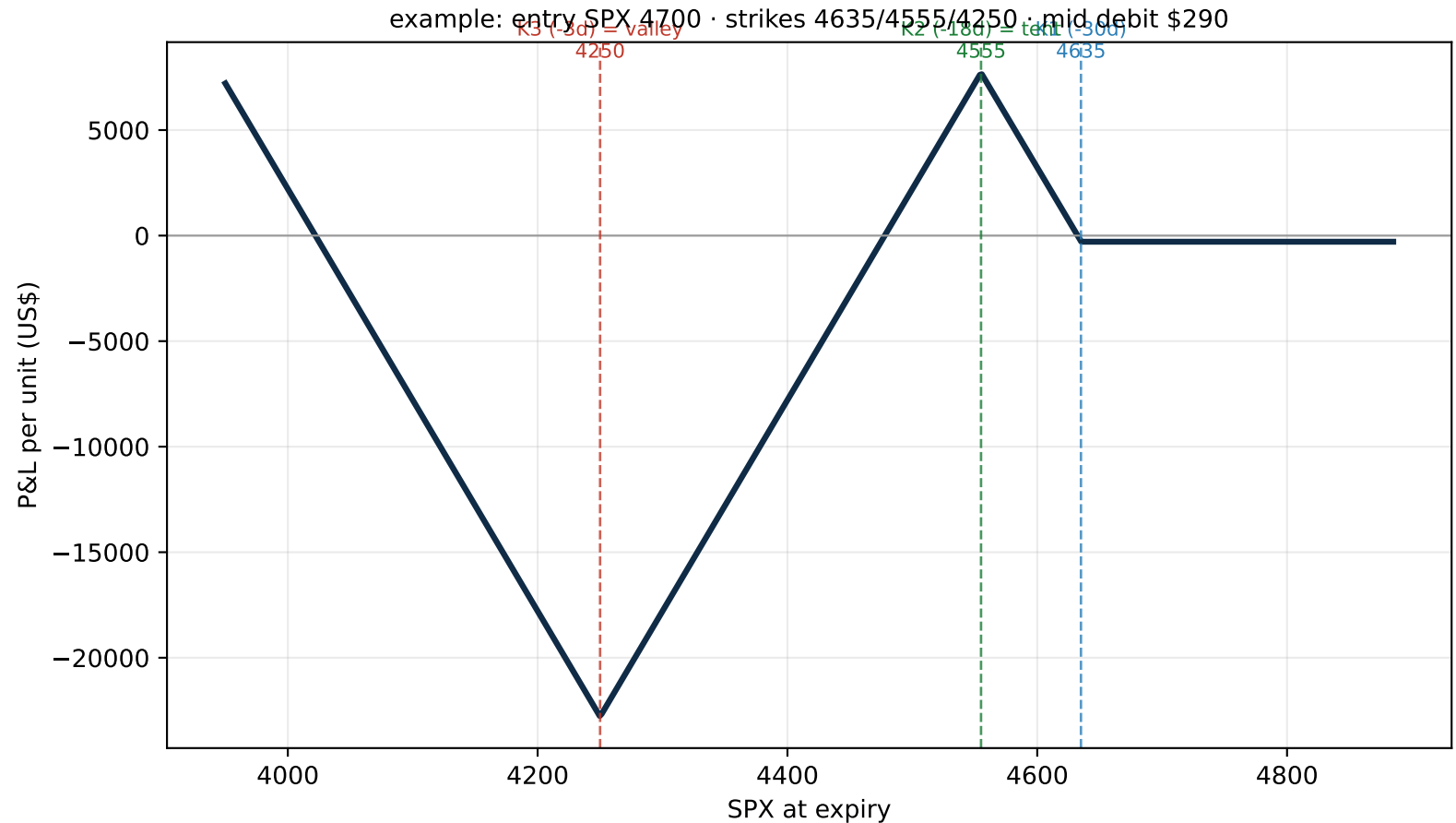
21-28 DTE positive (\$+66,642 / \$+99,348 at a 25% fill), 60 DTE not.

3. Holding to expiry is the WORST exit — a 'valley' reforms in the last days (CZ was right).
4. Real risk = a directional crash that lands in the valley (e.g. the 2025 tariff crash).

All P&L at MID pricing. Methodology and the leg-vs-combo point are explained on pages 2-3.

2. How the structure works

expiry payoff of one unit — where it makes and loses money



Reading the payoff (left to right):

- Above K1 -> all puts expire worthless: you lose only the small debit paid.
- Around K2 -> the 'tent': maximum profit if the market drifts down moderately.
- Around K3 -> the 'valley': the defined MAXIMUM LOSS (a crash that stops right here hurts).
- Far below K3 -> the +2 deep puts dominate again: convex payoff in a deep crash (the tail).

Entry: every Friday 10:00 ET, strikes chosen by delta (-30 / -18 / -3). Defined risk, no naked legs.

3. How we backtested it (and why it's trustworthy)

methodology — full transparency

We deliberately removed every source of fake P&L. What CZ should know:

ENGINE — synthetic tracking.

We subscribe to the real option chain and compute P&L analytically from the actual bid/ask of each leg. We do NOT route orders through QuantConnect's fill/margin engine, which fabricated phantom P&L on multi-leg structures. So the numbers reflect real quoted prices, nothing invented.

SETTLEMENT — exact, no spread.

SPX options are European, cash-settled. At expiry the position settles at the official intrinsic value — there is NO exit spread. Verified on page 8 (theoretical payoff == recorded P&L).

PRICING — mid is the standard; we also stress the fill.

Entry, marking and exits all use MID consistently. Because settlement has no spread, the ONLY thing separating 'mid' from 'bid/ask' on a hold is the ENTRY price of the 5 legs.

LEG-BY-LEG vs COMBO — the most important nuance for CZ.

Our pessimistic case sums each leg at its OWN bid/ask (as if you sent 5 separate market orders). A butterfly is never traded that way — it is sent as ONE combo at a single net limit. The combo's net spread is far tighter than the sum of the legs (the legs' risks offset; the MM hedges the package). Since mids are additive, our 'mid' number IS the combo's net mid — the price you would actually aim for. So reality sits close to mid, NOT at the leg-by-leg worst case.

VERIFIED at minute resolution.

We suspected the wide tail spread was a stale hourly-quote artifact. We re-ran at minute data: the entry spread is IDENTICAL (1.000x). It is the real far-OTM tail market, not a data glitch.

4. Fill-price scenarios — net P&L and Profit Factor (hold to expiry)

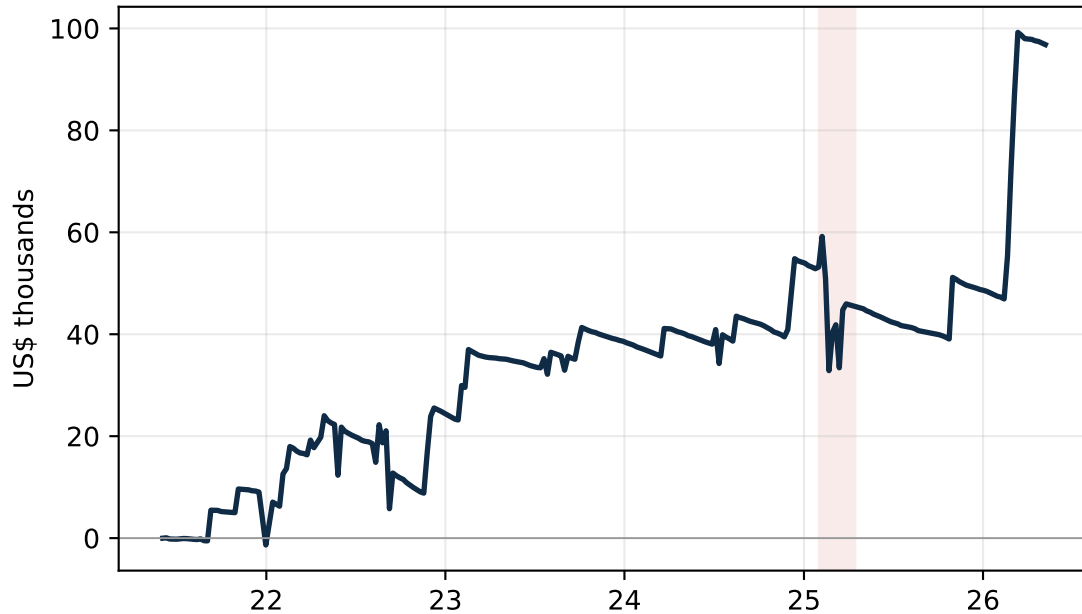
same trades, four entry-fill assumptions · realistic = combo limit order, near 25-50% · 5-year net

Horizon	Mid (optimistic)	25% spread (realistic)	50% spread	Leg-by-leg (worst, ref.)
21 DTE	\$+96,831 (PF 1.7)	\$+66,642 (PF 1.4)	\$+36,453 (PF 1.2)	\$-23,924 (PF 0.9)
28 DTE	\$+131,102 (PF 1.9)	\$+99,348 (PF 1.6)	\$+67,594 (PF 1.3)	\$+4,087 (PF 1.0)
45 DTE	\$+87,811 (PF 1.4)	\$+49,419 (PF 1.2)	\$+11,026 (PF 1.0)	\$-65,759 (PF 0.8)
60 DTE	\$+1,897 (PF 1.0)	\$-47,219 (PF 0.8)	\$-96,336 (PF 0.7)	\$-194,568 (PF 0.5)
75 DTE	\$+337,792 (PF 3.6)	\$+291,396 (PF 2.7)	\$+244,999 (PF 2.2)	\$+152,207 (PF 1.5)
100 DTE	\$+465,179 (PF 5.5)	\$+415,997 (PF 4.0)	\$+366,814 (PF 3.1)	\$+268,449 (PF 2.1)
120 DTE	\$+410,079 (PF 4.9)	\$+356,459 (PF 3.4)	\$+302,839 (PF 2.6)	\$+195,599 (PF 1.7)

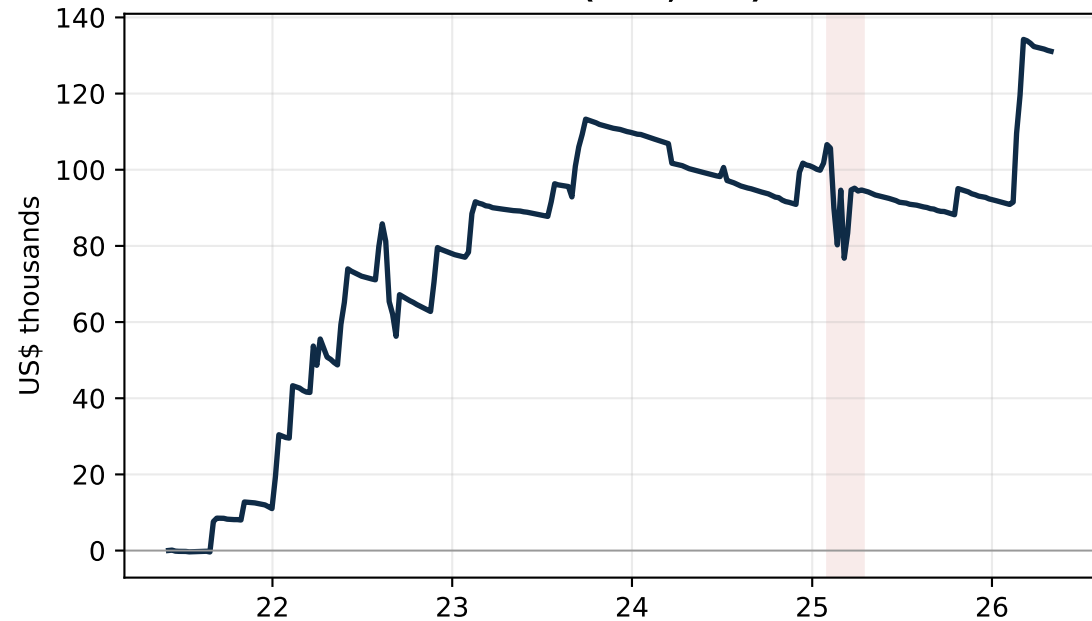
HOW TO READ: nobody buys at the ask and sells at the bid on every leg — the 'leg-by-leg' column is an unrealistic worst-case REFERENCE only. A butterfly is sent as ONE combo at a net limit, so the real fill sits between MID (optimistic) and ~25-50% of the spread. NOTE how the Profit Factor deflates with worse fills: e.g. 120 DTE goes from PF ~4.9 (mid) toward ~1.7 as the illiquid -3-delta tail spread is paid. The headline mid numbers are the CEILING, not the expectation; the honest read is the 25-50% columns. Short DTEs (21-28) stay positive; long DTEs keep a real but modest edge.

5. Cumulative P&L (mid) by horizon — full 5 years

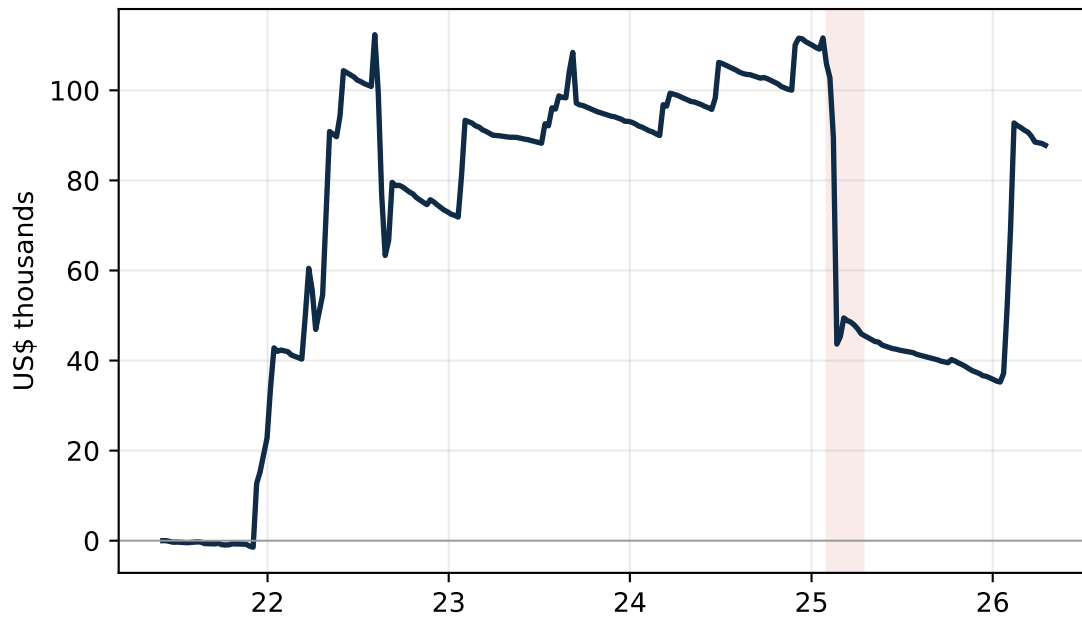
21 DTE (hold, mid)



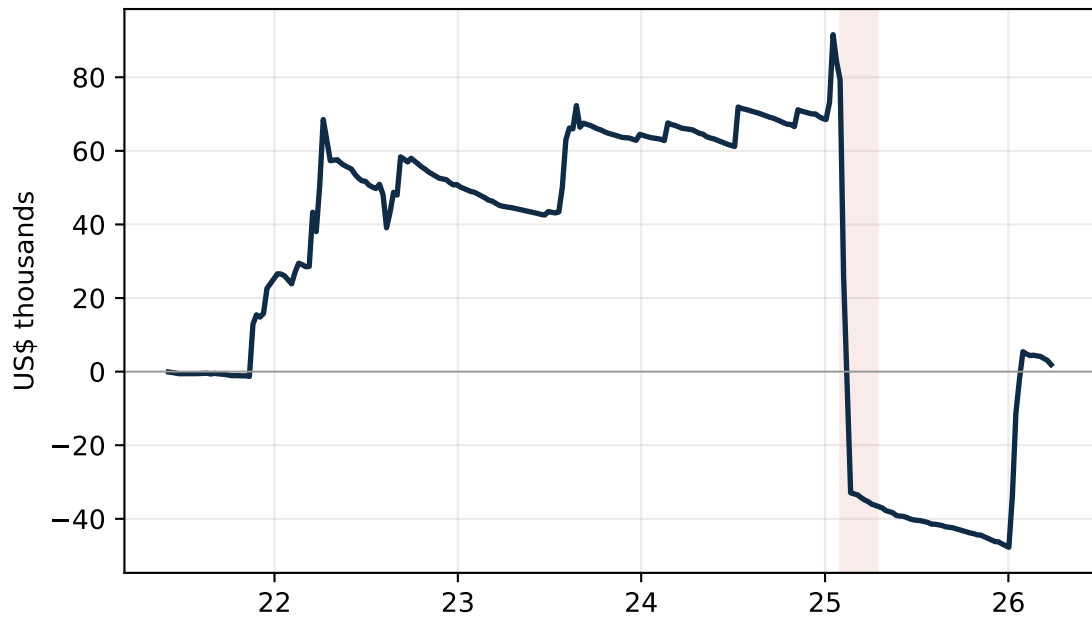
28 DTE (hold, mid)



45 DTE (hold, mid)



60 DTE (hold, mid)



Shaded band = the 2025 tariff crash. All curves are MID; net of the entry spread the slopes flatten (see page 4).

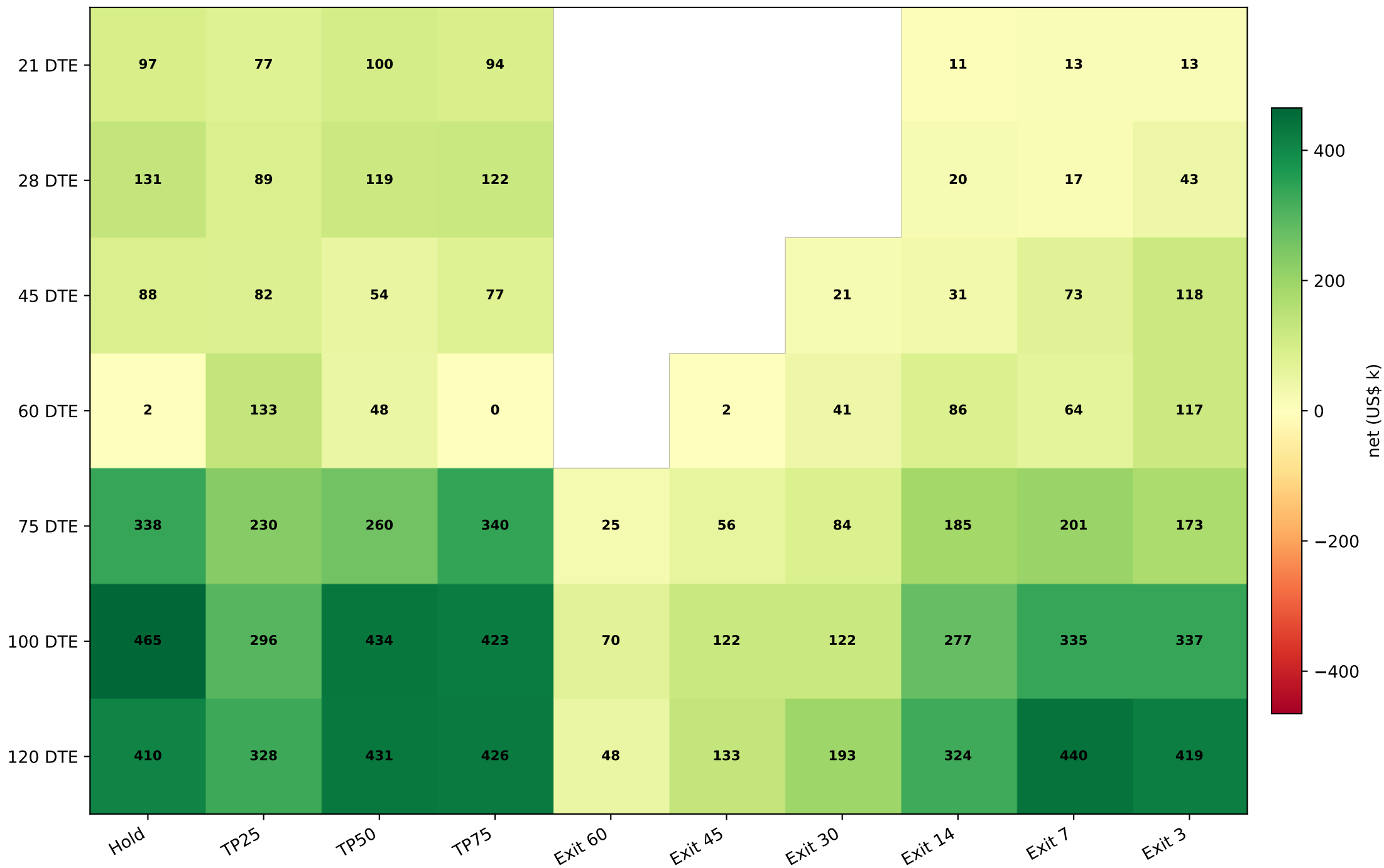
6. Exit rule x horizon — the full matrix (net P&L, mid)

every close rule x every horizon · 5-year net at mid

	Close rule	21 DTE	28 DTE	45 DTE	60 DTE	75 DTE	100 DTE	120 DTE
	Hold to expiry	\$+96,831	\$+131,102	\$+87,811	\$+1,897	\$+337,792	\$+465,179	\$+410,079
Exit at 30 DTE le	-	-	\$+21,105	\$+40,810	\$+84,405	\$+122,392	\$+193,498	
Exit at 21 DTE le	-	\$+8,485	\$+18,008	\$+52,508	\$+114,665	\$+222,065	\$+307,790	
Exit at 14 DTE le	\$+10,600	\$+20,035	\$+31,242	\$+85,928	\$+185,470	\$+276,790	\$+324,048	
Exit at 10 DTE le	\$+17,725	\$+22,250	\$+64,602	\$+81,588	\$+163,665	\$+274,118	\$+372,375	
Exit at 7 DTE le	\$+13,490	\$+17,372	\$+73,268	\$+63,975	\$+200,938	\$+334,642	\$+440,438	
Exit at 5 DTE le	\$+22,955	\$+20,355	\$+60,300	\$+77,615	\$+191,208	\$+386,388	\$+446,100	
Exit at 3 DTE le	\$+12,508	\$+42,535	\$+118,272	\$+117,478	\$+172,932	\$+337,478	\$+418,882	

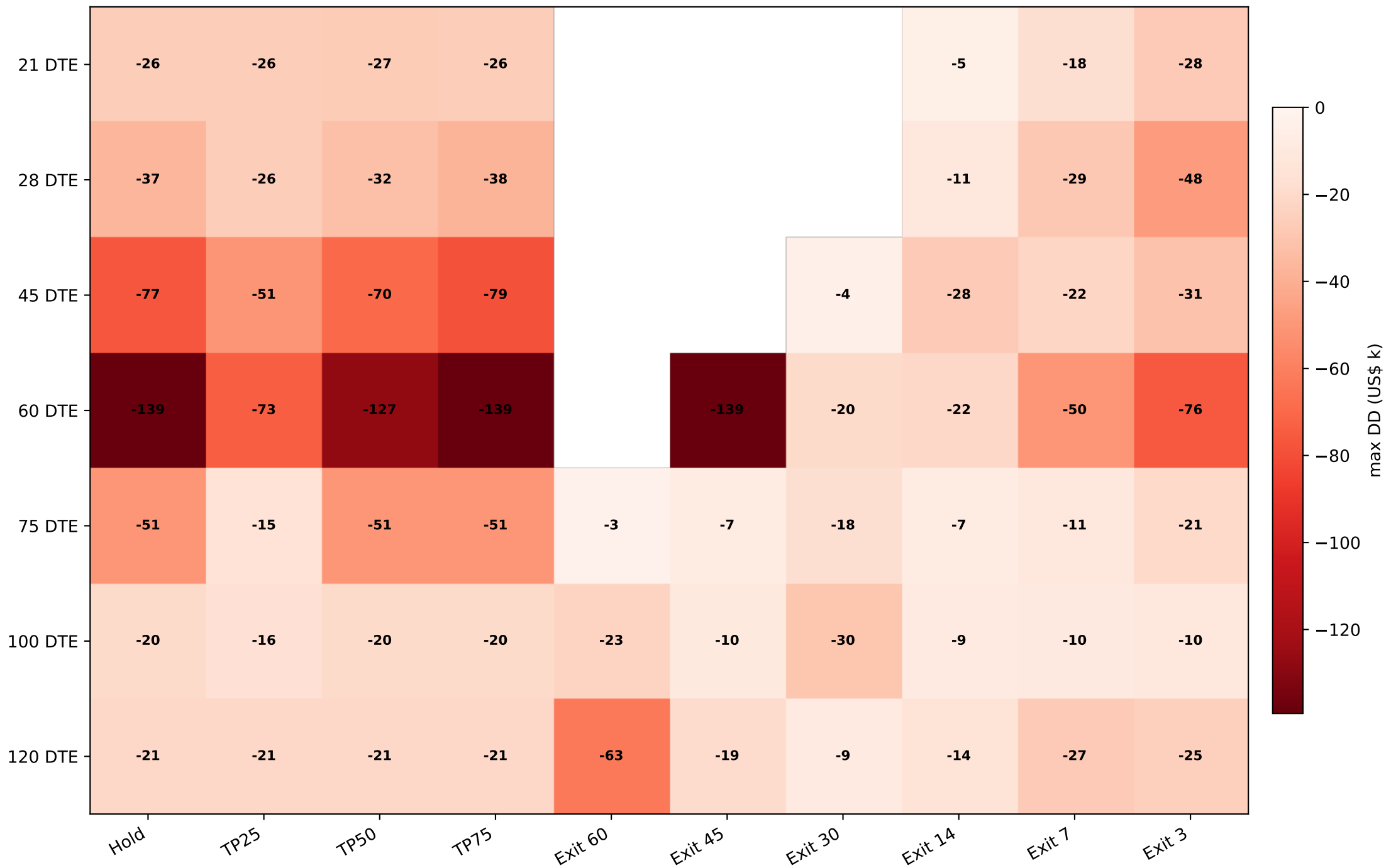
KEY INSIGHT — the horizon ranking FLIPS between holding and exiting early: HELD TO EXPIRY, short horizons win (28 DTE best; 60 DTE collapses to +\$2k, its valley is deepest at expiry). With EARLY EXIT (how PL5 is actually traded) LONG horizons win (45/60 DTE; e.g. exit 3 days before expiry: 45 DTE +\$118k vs 21 DTE +\$13k). Longer DTE gives the downside tent more time to form, and leaving before the final days avoids the valley reforming (60 DTE: +\$2k held vs +\$117k if exited 3 days early). All at mid; realistic combo fill scales every cell down similarly, so the relative ranking holds.

6b. Net P&L heatmap — horizon x close-rule (US\$ thousands, mid)



ner = more profit. Mid pricing (ceiling). Long DTEs glow on absolute \$ — but see the margin page: that P&L rides \$1-2M of capital. Best risk-adjusted cells are the short-DTE TP e

6c. Drawdown heatmap — horizon x close-rule (closed-trade equity, US\$ k, mid)



se to COMPARE close-rules within a horizon (TP exits, esp. short DTE, cut it most). The absolute mark-to-market risk (open positions) is on page 6d; this realized view understates

6d. Risk-adjusted metrics by horizon (hold, mid)

maxDD = aggregate MARK-TO-MARKET drawdown (real, incl. open positions) · Sharpe on monthly returns / avg capital

Horizon	Net	Max drawdown (MTM)	Sharpe	Calmar
21 DTE	\$+96,831	\$-27,460	0.68	0.71
28 DTE	\$+131,102	\$-52,362	0.95	0.51
45 DTE	\$+87,811	\$-93,152	0.40	0.19
60 DTE	\$+1,897	\$-159,415	0.01	0.00
75 DTE	\$+337,792	\$-70,211	1.07	1.01
100 DTE	\$+465,179	\$-49,150	1.37	2.03
120 DTE	\$+410,079	\$-79,290	1.25	1.11

MAX DRAWDOWN here is the aggregate MARK-TO-MARKET drawdown — the worst peak-to-trough of the whole open book, not just closed trades (the closed-trade curve badly understates risk on long DTEs, which sit open for months). It is a HISTORICAL drawdown (what happened in 5y), distinct from the margin page's THEORETICAL worst case (\$1-2M). Note it samples ~17 points/trade, so it is a floor — continuous marking would be somewhat deeper. SHARPE is modest because PL5 has a low win rate with fat crash-tail winners (very skewed monthly P&L = high volatility), so Sharpe under-rewards a convex payoff; read it together with Calmar (net/|maxDD|), which is strong for 75-120 DTE. All hold/mid (ceiling); realistic fills lower it.

7. A real worst case — the 2025 tariff crash (60 DTE)

trade 2025-02-07 -> 2025-04-04 · entry SPX 6099 · strikes 5975/5790/5050

Exit point	SPX that day	P&L (mid)
14 DTE left (2025-03-21)	5668	\$+475
10 DTE left (2025-03-25)	5777	\$+5,120
7 DTE left (2025-03-28)	5581	\$+4,065
3 DTE left (2025-04-01)	5633	\$-1,765
SETTLE (2025-04-04)	5075	\$-53,541

SPX collapsed in the final days to 5075, right at K3 (5050) = the valley = max loss. A holder took \$-53,541; exiting a week earlier was near flat. This is the genuine risk of PL5: a directional crash that lands in the valley. The loss is defined, but it is large and, with overlapping weekly entries, can cluster. This is real risk, at mid — not an execution artifact.

8. Margin & capital efficiency (CZ uses IB Portfolio Margin)

weekly entries stack: a 120 DTE holds ~18 packages at once · margin shown on the Reg-T ruler

Horizon	Max concurrent	Net (mid)	Reg-T margin (peak)	Avg margin	Return / margin
21 DTE	3	\$+96,831	\$169,440	\$74,526	57%
28 DTE	4	\$+131,102	\$252,480	\$123,571	52%
45 DTE	7	\$+87,811	\$463,030	\$243,214	19%
60 DTE	10	\$+1,897	\$799,040	\$411,963	0%
75 DTE	12	\$+337,792	\$1,098,440	\$586,621	31%
100 DTE	15	\$+465,179	\$1,526,540	\$880,394	30%
120 DTE	18	\$+410,079	\$1,947,330	\$1,144,411	21%

Margin here = Reg-T (sum of each package's defined max-loss, the valley) — the conservative ruler / ceiling. THE READ: long DTEs post a huge absolute P&L only by tying up far more capital — a 120 DTE needs ~\$1.9M peak across ~18 stacked positions and returns only ~21% on it, vs 21-28 DTE at ~52-57%. SHORT DTEs are ~2x more capital-efficient. CZ trades IB PORTFOLIO MARGIN, where the requirement is LOWER than Reg-T (PM stress-tests the book over a short move with time value intact, rather than summing every worst case) — but the exact figure comes from IB's TIMS model and we won't guess it. Crucially, PM trims the dollar amount but NOT the ranking: the efficiency gap (short >> long) holds on any ruler. Size by capital, not by the headline P&L. (All hold-to-expiry, mid; long-DTE win rate ~26-30%.)

9. Data verification

methodological rigor

Why these numbers are real, not fabricated:

- 1) Settlement = exact intrinsic at the SPX official close (cash-settled, European) — no spread, no phantom fills. Check on the 3 worst 60 DTE trades (theoretical payoff vs recorded hold):
 - 2025-02-07: SPX 5075 -> theoretical payoff \$-52,976 (recorded hold \$-53,541)
 - 2025-02-21: SPX 5361 -> theoretical payoff \$-28,880 (recorded hold \$-29,455)
 - 2025-02-14: SPX 5361 -> theoretical payoff \$-28,880 (recorded hold \$-29,245)
- 2) Strikes verified per trade (chosen at -30 / -18 / -3 delta).
- 3) MID pricing applied consistently across entry, marking and every exit rule.
- 4) Entry spread VERIFIED at minute resolution = identical to hourly (1.000x): the tail spread is the real market, not a stale-quote artifact (so the worst case is a legitimate bound).
- 5) Every trade is auditable in the app (date, strikes, DTE, debit, VIX, spot, exits, P&L).

10. Verdict

honest conclusion

PL5 is NOT a 'hold to expiry' trade — the valley at the end destroys it. And it is NOT a market-order trade — it must be worked as a combo at a limit price.

Realistic execution (combo fill, ~25-50% of the leg-by-leg spread), 5-year net P&L:

- 21 DTE: \$+66,642 / \$+36,453 -> positive, best return-on-margin (~57%)
- 28 DTE: \$+99,348 / \$+67,594 -> best net, ~52% on margin
- 45 DTE: \$+49,419 / \$+11,026 -> marginal
- 60 DTE: \$-47,219 / \$-96,336 -> negative
- 75-120 DTE (CZ request): huge ABSOLUTE P&L (+\$338k to +\$465k hold/mid) BUT need \$1.1-1.9M margin (12-18 stacked positions) -> only ~21-31% on capital, and low win rate (~26-30%).

Honest caveats:

- Execution is THE decisive factor (5 legs, illiquid -3d tail). Fill discipline = the edge.
- Long DTEs win on headline \$ only by tying up far more capital — short DTEs are 2x more capital-efficient (see page 8). Size by margin, not by absolute P&L.
- Real tail risk: a crash into the valley = a large, defined loss; clusters with overlapping entries.

Suggested next step: forward-test 28 DTE (best risk-adjusted) with early exit/TP + a concurrency cap, and measure the REAL combo fill on the tail — the single factor that decides this strategy.